Report for exercise BE 01 from group 01

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AI Disclaimer
OpenAI ChatGPT 5.5 was used in the creation of this report. Specifically:

- discussing code results
- latex formulas
- latex tables
- debugging numpy issues
- creating code for plots

Report on task 1.1 Mean and Variance Estimation

In this task, the sample mean and sample variance of a given dataset were computed using both a manual Python implementation and NumPy's built-in statistical routines.

Given samples $\{X_i\}_{i=1}^N$, the sample mean is defined as

$$\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i.$$

The worksheet defines the unbiased sample variance estimator as

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(X_i - \bar{X}\right)^2.$$

This formulation utilizes the Bessel correction term with normalization factor $N-1$. The correction compensates for the loss of one degree of freedom introduced by estimating the sample mean from the same dataset. Consequently, the biased estimator based on normalization by N systematically underestimates the true variance for finite sample sizes.

The Python implementation explicitly follows this definition and therefore divides by $N-1$. In contrast, NumPy's default variance implementation uses `ddof=0`, corresponding to the biased population variance estimator

$$\frac{1}{N}\sum_{i=1}^{N}\left(X_i - \bar{X}\right)^2.$$

To obtain the unbiased estimator with NumPy, the parameter `ddof=1` must be specified explicitly.

The obtained numerical results are summarized in Table 1.

Implementation	Mean	Variance
Python implementation	1.647059	0.148897
NumPy (`ddof=0`)	1.647059	0.140138
NumPy (`ddof=1`)	1.647059	0.148897

Table 1: Comparison of mean and variance estimators for the grade dataset.

The means agree exactly because all implementations compute the same sample mean estimator. Likewise, the manually implemented variance agrees exactly with NumPy for `ddof=1`, confirming that both implementations compute the same unbiased estimator.

For the considered dataset with $N = 17$ samples, the numerical difference between the biased and unbiased variance estimators remains noticeable. The corresponding Bessel correction factor is

$$\frac{N}{N-1} = \frac{17}{16} = 1.0625.$$

Consequently, the biased estimator underestimates the unbiased variance by approximately 5.88%, while the unbiased estimator is approximately 6.25% larger than the biased estimator.

In later tasks of this report, covariance estimation, Monte Carlo convergence, and uncertainty propagation rely directly on empirical variance estimators. In these settings, estimator bias affects the quality of uncertainty estimates and therefore the interpretation of convergence behavior.

For this reason, the unbiased estimator with `ddof=1` was used consistently throughout the remainder of the report whenever empirical variances or covariance matrices were computed.

Report on task 1.2 Multivariate Mean and Covariance Estimation

In this task, samples from a multivariate Gaussian distribution were generated in order to estimate the mean vector and covariance matrix empirically. The target distribution was defined by

$$\mu = \begin{bmatrix} -0.4 \\ 1.1 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 2 & 0.4 \\ 0.4 & 1 \end{bmatrix}.$$

For increasing sample sizes $N \in \{10, 100, 1000, 10000\}$, Monte Carlo samples were drawn and empirical estimators for the mean and covariance were computed.

The empirical covariance estimator was implemented as

$$\hat{\Sigma} = \frac{1}{N-1} \sum_{i=1}^{N} (X_i - \bar{X})(X_i - \bar{X})^T,$$

corresponding to the unbiased sample covariance estimator. Consistently with Task 1.1, the normalization by $N-1$ was used throughout the report in order to avoid the systematic underestimation of variance arising from the biased estimator.

The obtained numerical estimates are summarized in Table 2.

N	Estimated mean	Estimated covariance
10	$\begin{bmatrix} -0.5811 \\ 0.7875 \end{bmatrix}$	$\begin{bmatrix} 1.6687 & -0.1651 \\ -0.1651 & 0.4709 \end{bmatrix}$
100	$\begin{bmatrix} -0.3655 \\ 1.0668 \end{bmatrix}$	$\begin{bmatrix} 1.7596 & 0.2305 \\ 0.2305 & 0.6503 \end{bmatrix}$
1000	$\begin{bmatrix} -0.2895 \\ 1.1006 \end{bmatrix}$	$\begin{bmatrix} 2.0179 & 0.3210 \\ 0.3210 & 0.9675 \end{bmatrix}$
10000	$\begin{bmatrix} -0.4009 \\ 1.1116 \end{bmatrix}$	$\begin{bmatrix} 1.9879 & 0.3641 \\ 0.3641 & 1.0091 \end{bmatrix}$

Table 2: Monte Carlo estimates of the mean vector and covariance matrix for increasing sample sizes.

Figure 1 shows the convergence behavior of the estimation errors and the RMSE on a log-log scale.

As expected from Monte Carlo theory, the estimation errors decrease with increasing sample size. The RMSE approximately follows the theoretical convergence rate

$$\mathcal{O}(N^{-1/2}),$$

corresponding to the standard Monte Carlo convergence behavior derived in the lecture.

The empirical mean and covariance estimates approach the exact values as the number of samples increases. The mean estimator converges more smoothly than the covariance estimator. This behavior is expected because covariance estimation depends on second-order moments and therefore exhibits larger sampling variability.

Interestingly, the mean error temporarily increases at $N = 1000$. Although this may initially appear counterintuitive, it is a natural consequence of the stochastic nature of Monte Carlo sampling. The Monte Carlo estimator is itself random, and finite-sample fluctuations may therefore occasionally lead to larger errors for intermediate sample sizes. Since only a single realization is shown for each sample size, these fluctuations remain visible in the plot. Nevertheless, the overall convergence trend still follows the theoretical Monte Carlo rate.

In this realization, the off-diagonal covariance error remains larger than the diagonal covariance error for most sample sizes, particularly for larger N. Estimating correlations between random variables is generally more difficult than estimating marginal variances because the estimator depends simultaneously on fluctuations in multiple dimensions.

The RMSE curve behaves substantially more smoothly than the exact error because the RMSE represents an expected statistical error averaged over hypothetical repetitions of the experiment, whereas the exact error corresponds only to one particular realization of the random samples.

The convergence of the covariance estimator is visibly slower and more irregular than that of the mean estimator. In addition, the RMSE formula derived for the Monte Carlo mean estimator cannot directly be transferred to the covariance estimator. The covariance estimator possesses a significantly more complicated variance structure because it depends quadratically on the samples and additionally on the estimated sample mean itself.

These observations become particularly relevant in the later tasks of this report, where covariance information directly influences uncertainty propagation and the effectiveness of variance reduction techniques.

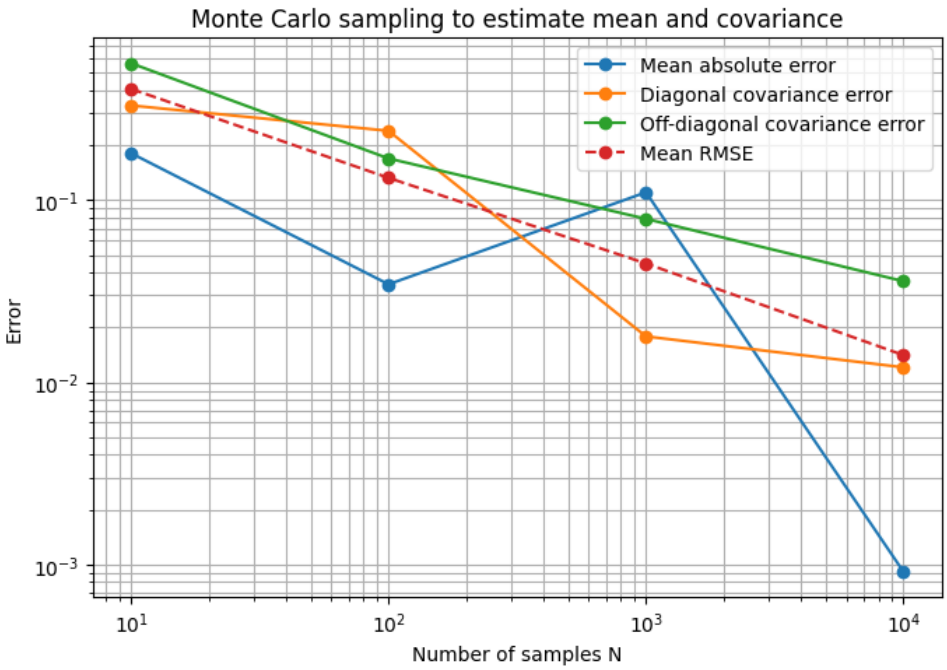

Figure 1: Convergence of the mean and covariance estimation errors for increasing sample sizes.

Report on task 2 Monte Carlo Integration

In this task, Monte Carlo integration was applied to approximate the integral

$$I = \int_a^b \sin(x)\, dx.$$

The analytical reference solution is given by

$$I = \cos(a) - \cos(b).$$

The integral was estimated using independent samples drawn from uniform distributions. Two approaches were considered:

- direct sampling from the interval $[a, b]$,
- transformation of samples from the reference interval $[0, 1]$ to $[a, b]$.

For the transformed approach, samples were mapped according to

$$x = a + (b - a)u, \qquad u \sim \mathcal{U}(0, 1).$$

The Monte Carlo estimator of the integral is

$$\hat{I} = (b - a)\frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

The corresponding RMSE was estimated empirically from the sampled function evaluations.

The obtained numerical results are summarized in Tables 3–5.

N	Estimate	Exact error	RMSE
10	0.47581552	0.01611782	0.08636807
100	0.43302626	0.02667143	0.02570408
1000	0.45089187	0.00880583	0.00794286
10000	0.45483267	0.00486502	0.00247115

Table 3: Monte Carlo integration on $[0, 1]$. Exact integral: 0.45969769.

N	Estimate	Exact error	RMSE
10	0.15253903	0.08495776	0.35731470
100	0.34015134	0.10265455	0.10616370
1000	0.27111060	0.03361382	0.03304667
10000	0.25874447	0.02124769	0.01030407

Table 4: Direct Monte Carlo integration on $[2, 4]$. Exact integral: 0.23749678.

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10	0.15253903	0.08495776	0.35731470
100	0.34015134	0.10265455	0.10616370
1000	0.27111060	0.03361382	0.03304667
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Table 5: Monte Carlo integration on $[2, 4]$ using transformed samples from $[0, 1]$. Exact integral: 0.23749678.

Figures 2–4 show the convergence behavior for increasing sample sizes.

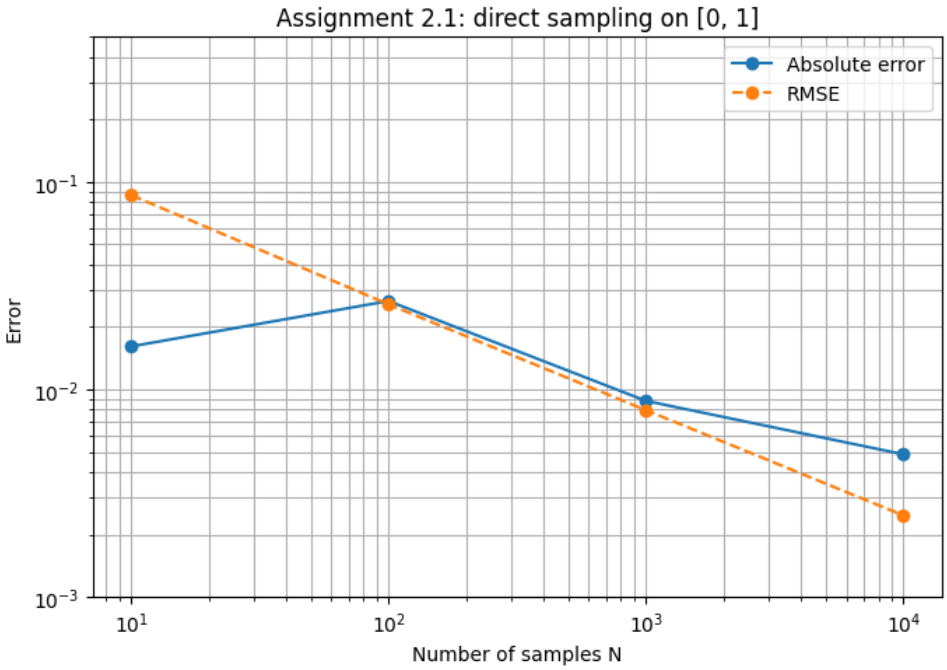

Figure 2: Monte Carlo integration on $[0, 1]$.

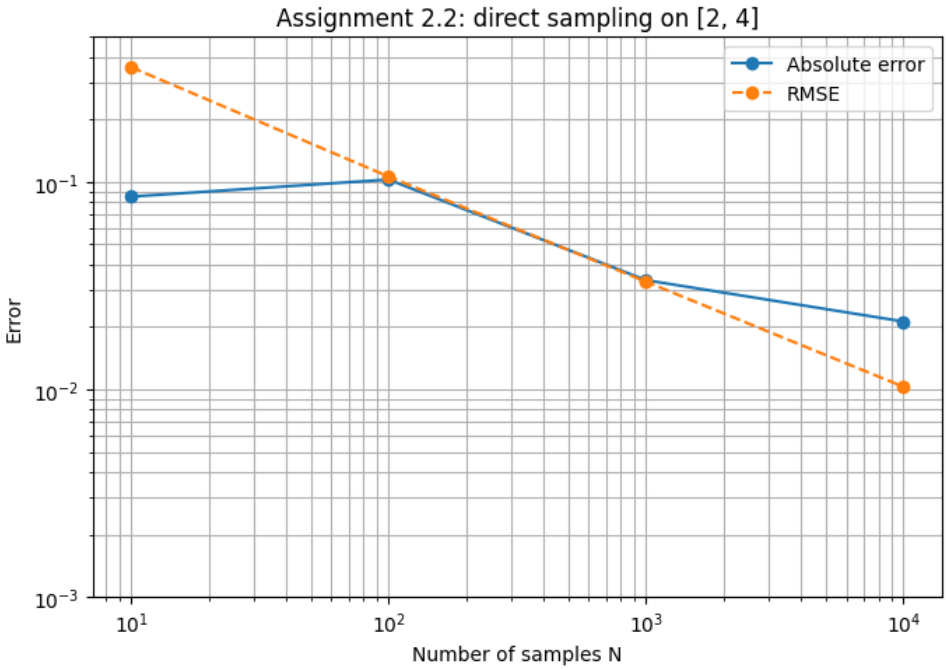

Figure 3: Direct Monte Carlo integration on $[2, 4]$.

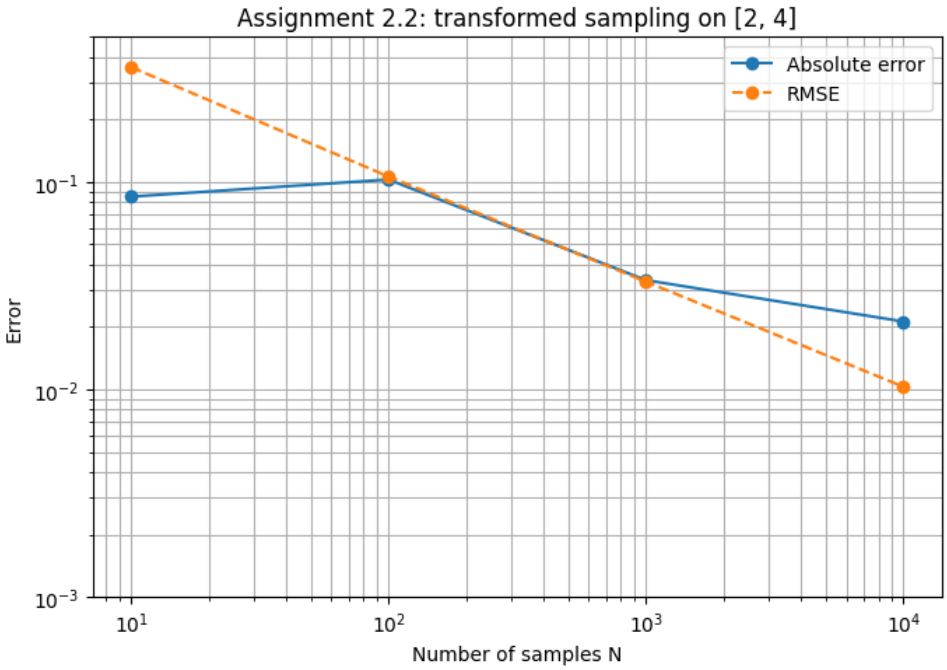

Figure 4: Monte Carlo integration on $[2, 4]$ using transformed samples from $[0, 1]$.

The numerical results confirm the characteristic Monte Carlo convergence rate

$$\mathcal{O}(N^{-1/2}),$$

independently of the integration interval. Furthermore, the transformed and directly sampled estimators on $[2, 4]$ exhibit identical convergence behavior, as expected theoretically since both approaches generate samples from the same target distribution.

The RMSE provides an estimate of the expected statistical error of the Monte Carlo estimator, whereas the exact error corresponds to one particular realization of the random sampling process. Consequently, the exact error does not decrease monotonically and may occasionally be smaller or larger than the RMSE estimate. This behavior is clearly visible in the numerical results and reflects the stochastic nature of Monte Carlo integration.

For Assignment 2.2, the RMSE retains the same statistical interpretation as in Assignment 2.1 because the transformed samples still represent realizations of a uniform random variable on $[2, 4]$. The affine transformation changes only the sampling domain but does not alter the underlying Monte Carlo averaging principle.

Compared to Task 1, this task highlights that Monte Carlo estimators are fundamentally expectation estimators. The uncertainty of the integral approximation is therefore directly connected to the variance of the sampled function values.

Report on task 3 Variance Reduction Methods

In this task, variance reduction techniques for Monte Carlo integration were investigated using the target function

$$f(x) = e^x, \qquad x \in [0, 1].$$

The exact value of the integral is

$$I = \int_0^1 e^x \, dx = e - 1.$$

Three different approaches were compared:

- standard Monte Carlo integration,
- control variates,
- importance sampling.

The obtained absolute errors are summarized in Table 6.

N	MC	CV h_1	CV h_2	CV h_3	IS $\beta(5,1)$	IS $\beta(0.5,0.5)$
10	3.8746	0.5176	0.5176	0.0907	52.6304	2.0013
100	4.6221	0.3523	0.3523	0.1228	22.3869	8.3207
1000	1.4795	0.1610	0.1610	0.0572	31.8954	3.2457
10000	1.0349	0.0500	0.0500	0.0124	21.0658	0.1706

Table 6: Absolute integration errors for standard Monte Carlo, control variates, and importance sampling. Values are scaled by 10^{-2}, rounded to 4 decimals.

Figure 5 shows the absolute integration errors for increasing sample sizes.

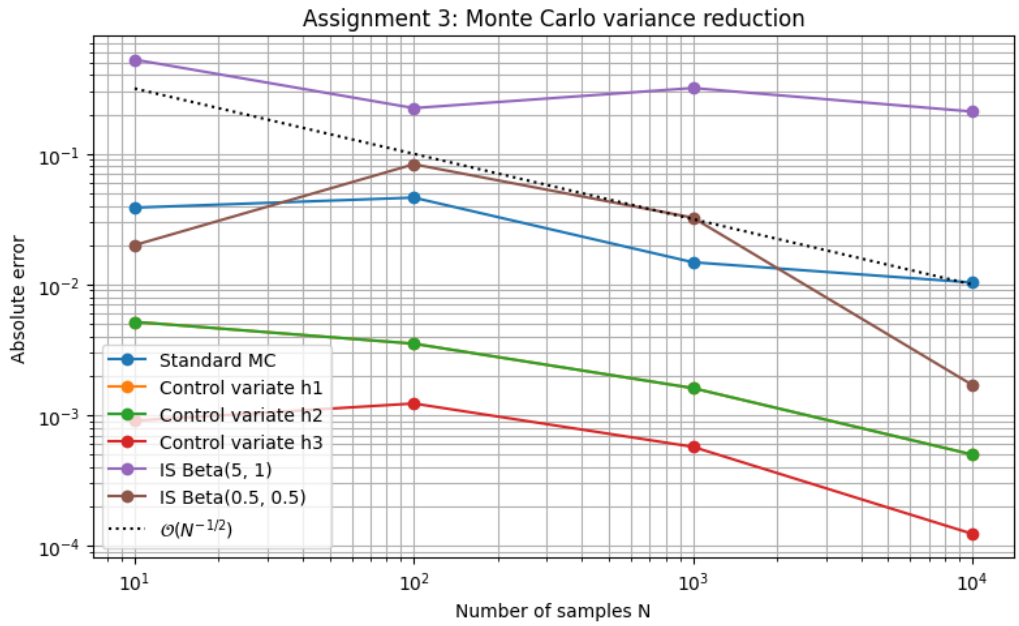

Figure 5: Comparison of standard Monte Carlo, control variates, and importance sampling.

As expected from Monte Carlo theory, the standard Monte Carlo estimator exhibits the characteristic convergence behavior

$$\mathcal{O}(N^{-1/2}),$$

which is limited by the variance of the sampled integrand values.

For the control variate approach, the auxiliary functions

$$h_1(x) = x, \qquad h_2(x) = 1 + x, \qquad h_3(x) = 1 + x + \frac{x^2}{2}$$

were considered.

Control variates reduce the variance of the estimator by exploiting the correlation between the target function and an auxiliary function with known expectation. The closer the auxiliary function resembles the integrand, the stronger the variance reduction becomes.

The results show that h_1 and h_2 produce identical convergence behavior. This is expected because both functions differ only by a constant shift, which does not affect the variance reduction mechanism of the control variate estimator.

The best performance was achieved using h_3. This can be explained by the Taylor expansion

$$e^x = 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3),$$

showing that h_3 closely approximates the target integrand. Consequently, the correlation between the target function and the control variate becomes very strong, leading to substantial variance reduction.

For importance sampling, proposal distributions based on Beta distributions were investigated. Importance sampling aims to reduce variance by drawing samples preferentially from regions that contribute strongly to the integral. Ideally, the proposal distribution should resemble the shape of the integrand.

The proposal $\text{Beta}(5, 1)$ produced comparatively poor performance because it oversamples the region near $x = 1$, leading to highly uneven importance weights and therefore increased estimator variance. In contrast, $\text{Beta}(0.5, 0.5)$ produced substantially smaller errors than $\text{Beta}(5, 1)$, although its convergence behavior remained less stable than that of the control variate estimators.

Overall, the results demonstrate that appropriately chosen variance reduction techniques can significantly outperform standard Monte Carlo integration while preserving estimator consistency.

Report on task 4 Forward Uncertainty Propagation

In this task, uncertainty propagation was investigated for a damped harmonic oscillator with uncertain forcing frequency ω. Instead of considering a deterministic input parameter, the forcing frequency was modeled as a uniformly distributed random variable, leading to a stochastic system response.

The quantity of interest was the oscillator displacement at final time $t = 10$,

$$y(10),$$

for which the mean and variance were estimated numerically.

A deterministic reference trajectory corresponding to the nominal parameter value was additionally computed for comparison. However, since the forcing frequency is uncertain, the deterministic trajectory alone is insufficient to characterize the full system behavior. Uncertainty quantification therefore requires the computation of statistical quantities such as the expectation and variance of the solution.

The Monte Carlo and quasi-Monte Carlo estimators were compared using increasing sample sizes

$$N \in \{10, 100, 1000, 10000\}.$$

As reference values, highly resolved Monte Carlo estimates based on 10^6 samples were used:

$$\mathbb{E}[y(10)] = -0.43893703,$$

$$\mathrm{Var}[y(10)] = 0.00019678.$$

The deterministic solution at $t = 10$ was

$$y_{\mathrm{det}}(10) = -0.45273580.$$

The obtained relative errors are summarized in Table 7.

Method and N	Mean error	Variance error
Monte Carlo, $N = 10$	0.13658	14.52802
Monte Carlo, $N = 100$	0.34559	10.35184
Monte Carlo, $N = 1000$	0.12062	3.71903
Monte Carlo, $N = 10000$	0.00244	0.93889
Quasi-Monte Carlo, $N = 10$	0.03556	2.76362
Quasi-Monte Carlo, $N = 100$	0.02104	2.98448
Quasi-Monte Carlo, $N = 1000$	0.00676	0.15923
Quasi-Monte Carlo, $N = 10000$	0.00650	0.11845

Table 7: Relative errors of Monte Carlo and quasi-Monte Carlo estimators for the mean and variance of $y(10)$. Values are scaled by 10^{-2} and rounded.

Figure 6 shows the convergence behavior of the relative errors, while Figure 7 illustrates several oscillator trajectories corresponding to randomly sampled forcing frequencies.

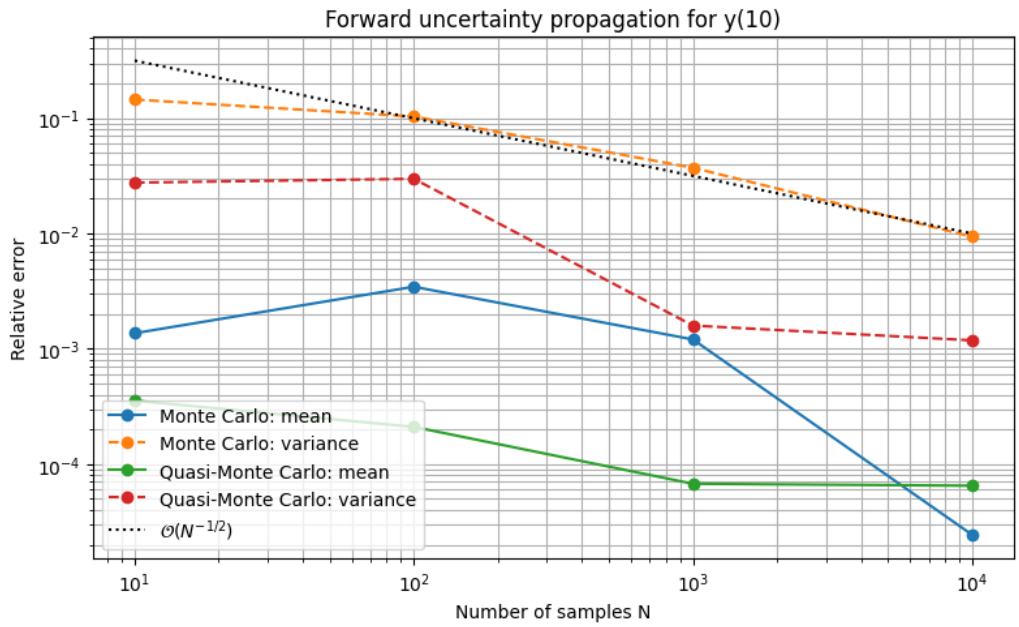

Figure 6: Relative errors of Monte Carlo and quasi-Monte Carlo estimators for increasing sample sizes.

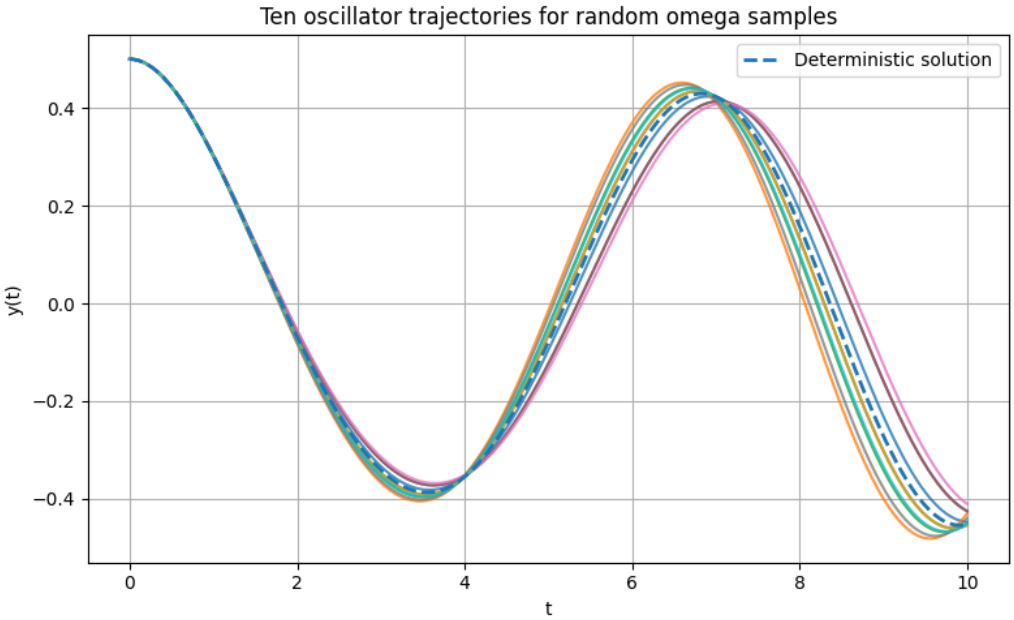

Figure 7: Oscillator trajectories for randomly sampled forcing frequencies together with the deterministic solution.

The numerical results confirm the characteristic convergence behavior expected from uncertainty propagation methods. Both Monte Carlo and quasi-Monte Carlo estimators show an overall decrease in error as the sample size increases, although the convergence is not strictly monotonic for every statistic due to finite-sample fluctuations. Overall, the quasi-Monte Carlo approach achieves substantially lower errors for most sample sizes, especially for the variance estimates. For the mean estimator, QMC performs better for $N = 10, 100, 1000$, while standard Monte Carlo happens to achieve the smallest mean error at $N = 10000$. This illustrates that individual error curves are still subject to finite-sample fluctuations.

This behavior can be explained by the improved space-filling properties of low-discrepancy sequences such as the Halton sequence used in the quasi-Monte Carlo implementation. In contrast to pseudo-random Monte Carlo sampling, quasi-Monte Carlo points cover the parameter space more uniformly, leading to reduced sampling variability and therefore faster practical convergence.

The variance estimates converge substantially more slowly than the mean estimates. This behavior is expected because variance estimation depends on second-order statistics and is therefore more sensitive to sampling fluctuations. The effect is particularly visible for standard Monte Carlo sampling, where the variance errors remain comparatively large even for moderate sample sizes.

The deterministic solution differs noticeably from the estimated mean response. This observation highlights a central concept of uncertainty quantification: for nonlinear systems, the expected response under uncertain inputs generally does not coincide with the deterministic solution obtained from nominal parameter values.

Overall, the results demonstrate that quasi-Monte Carlo methods can significantly improve the efficiency of forward uncertainty propagation problems while preserving estimator consistency.